DATA SCIENCE PROJECT REPORT

Supermarket Transaction Summary with Pandas

Exploratory data analysis • Business performance insights • Visual summaries

1,000	17	322,966.75	322.97
transaction records	original columns	total sales value	mean sales per record

1. Introduction

Retail transaction data captures how customers purchase products, which categories contribute to sales, how branches perform, and which payment options customers use. When this information is organized and summarized systematically, it can support clearer operational and commercial decisions. This project uses Python and Pandas to explore a supermarket transaction dataset and turn row-level records into concise business summaries.

2. Project Description

The notebook loads the supermarket analysis CSV into a Pandas DataFrame, inspects its structure, checks for missing and duplicate records, derives additional fields, and groups sales by branch, product line, payment method, day, and month. It also identifies high-value transactions and displays bar charts for selected comparisons. The work is descriptive analytics rather than a predictive machine-learning model.

Project focus

Convert transaction-level data into readable summaries that help compare sales performance and customer purchasing patterns. The analysis is designed to be reproducible and easy to extend as new transaction data becomes available.

3. Problem Statement

A raw transaction table contains many individual records, making it difficult to quickly compare branch performance, identify stronger product categories, understand payment preferences, or spot high-value purchases. The project addresses this problem by using Pandas aggregation, filtering, date extraction, and visualization to produce structured summaries from the transaction records.

4. Dataset Description and Usage

The notebook imports two files: **SuperMarket Analysis.csv** and **Supermarket Transactions.xlsx**. The Excel file loads as an empty DataFrame (0 rows and 0 columns) in the recorded run; the analysis therefore relies on the CSV file, which contains 1,000 records and 17 original columns.

The CSV includes invoice ID, branch, city, customer type, gender, product line, unit price, quantity, 5% tax, sales, date, time, payment, cost of goods sold (cogs), gross margin percentage, gross income, and customer rating. There is no separate prediction target because the notebook performs exploratory and descriptive analysis, not supervised learning.

Data preparation performed

• Inspected sample rows, shape, column types, and descriptive statistics.
• Checked missing values and duplicate rows; the recorded checks found zero missing values and zero duplicates.
• Applied drop_duplicates() and dropna() as defensive cleaning steps; these checks did not remove records in the observed dataset.
• Converted Date to datetime and derived Day and Month.
• Added Total = Unit price × Quantity for supplementary transaction-level analysis.

Data flow



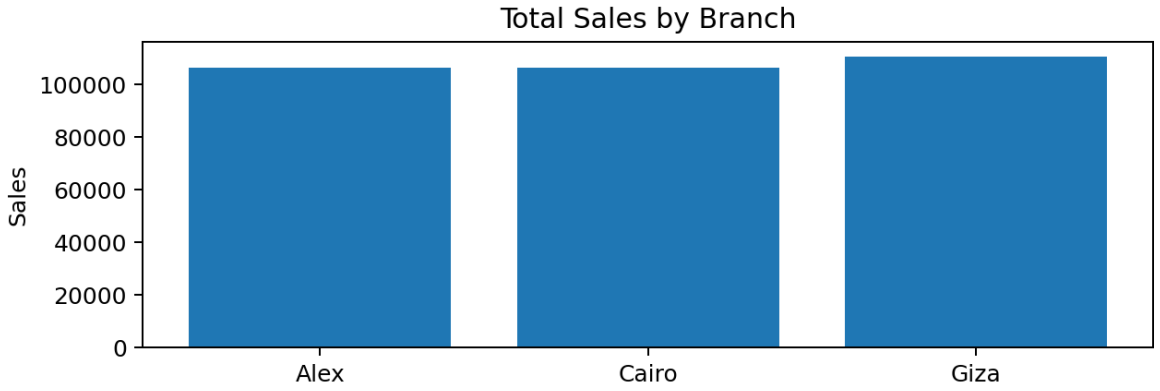
Note: The notebook also imports NumPy, but the visible analysis is primarily implemented with Pandas and Matplotlib-style plotting through DataFrame plot methods.

5. Workflow and Implementation

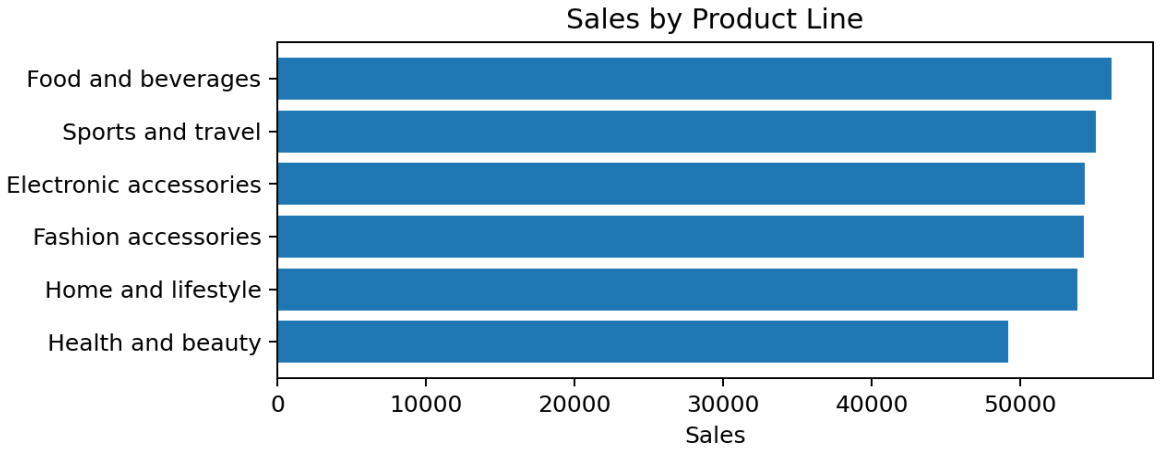
The implementation follows a straightforward exploratory analytics pipeline: load the files; inspect the dataset; validate completeness and uniqueness; create derived fields; aggregate metrics by relevant business dimensions; filter and sort records; and plot comparisons. Key Pandas operations include info(), describe(), isnull().sum(), duplicated(), drop_duplicates(), dropna(), to_datetime(), groupby(), agg(), value_counts(), and sort_values().

6. Results and Analysis

322,966.75	1,000	5.51
total sales	unique invoices	average quantity per record

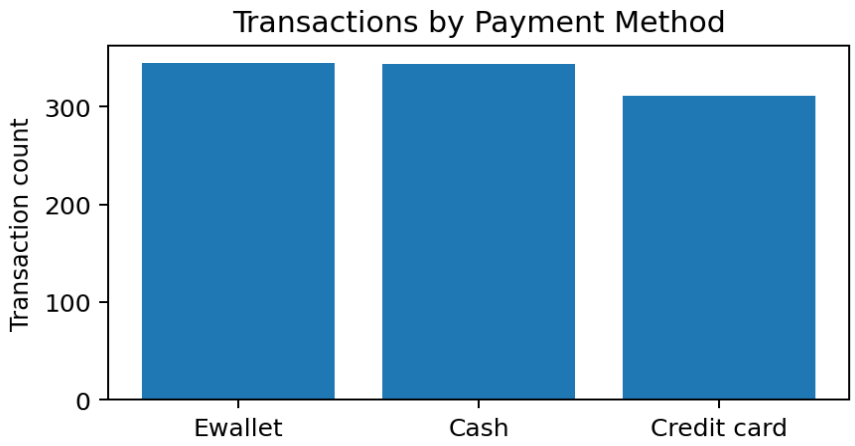


Branch comparison. Giza recorded the highest total sales at 110,568.71. Alex and Cairo were nearly tied at 106,200.37 and 106,197.67 respectively. Giza also had the highest average sales per transaction (337.10), compared with Alex (312.35) and Cairo (319.87).



Product lines. Food and beverages led sales at 56,144.84, followed by Sports and travel at 55,122.83. Health and beauty had the lowest total among the six product lines at 49,193.74. Values are the sums of the dataset's Sales column.

6. Results and Analysis (continued)



Payment methods. E-wallet was used in 345 transactions, cash in 344, and credit card in 311. E-wallet and cash usage were almost equal in this dataset; credit card usage was lower.

Additional findings

Measure	Observed result	Interpretation
Monthly sales	January 116,291.87; March 109,455.51; February 97,219.37	January was the strongest month in the recorded period.
Transaction activity by day	Saturday 164; Tuesday 158; Monday 125	Saturday had the most records; Monday had the fewest.
Highest individual sales	1,042.65	The largest transaction in the sorted top five was in Fashion accessories, at Giza, on 15 February 2019.
Quantity by product line	Electronic accessories 971 units; Food and beverages 952	Electronic accessories had the highest total quantity among product lines.

7. Conclusion

This project demonstrates how Pandas can turn supermarket transaction records into useful descriptive summaries. The notebook verified data completeness, added date-based fields and a calculated total, and compared branches, product lines, payment methods, and transaction timing. The results identify Giza as the highest-sales branch in the dataset, Food and beverages as the leading product line by sales, and e-wallet as the most frequently recorded payment method. These findings provide a practical starting point for retail performance review, while remaining descriptive rather than causal.

8. Future Scope

Future improvements could include a dashboard with interactive filters, customer-segment comparisons, weekday-by-hour analysis, profit-focused metrics, anomaly detection, and automated ingestion of updated files. Before extending the work, the Excel input should be checked because it was empty in the recorded run. Future findings should be revalidated against the refreshed data.

From transaction records to clearer retail decisions.